

Program : Diploma in Wood and Paper Technology	
Course Code : 5123C	Course Title: Process Control And Instrumentation (WP)
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on measurement of various parameters and instrumentation for processes in wood based industries.
- To provide knowledge of process control.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Fluid flow		Mechanical and electrical Engineering	3
Chemical processes and operations		Chemical Engineering	4
Pulp and paper manufacture		Pulp and paper Technology	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the importance of instrumentation and the static characteristics of instruments in wood based industries	15	Understanding
CO2	Describe the working principle of measuring instruments in process plants	14	Understanding
CO3	Outline the measuring instruments in paper industry	15	Understanding

CO4	Explain the modes of control action and the computer control system	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the importance of instrumentation and the static characteristics of instruments in wood based industries.		
M1.01	Explain the importance of Instrumentation in industries	1	Understanding
M1.02	Construct the instrumentation block diagram	1	Understanding
M1.03	Suggest the functions of measuring instruments	6	Understanding
M1.04	Analyze different static and dynamic characteristics of instruments	7	Understanding
Contents: Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error.			

CO2	Describe the working principle of measuring instruments in process plants		
M2.01	Explain the working principle of Temperature measuring instruments	3	Understanding
M2.02	Discuss the working principle of Pressure measuring instruments	3	Understanding
M2.03	Outline the working principle of flow and level measuring instruments	3	Understanding
M2.04	Illustrate the working principle of PH, viscosity and humidity measuring instruments	5	Understanding
	Series Test - I	1	

Contents:

Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical.

Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors.

Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters.

Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell.

PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter – Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type

Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer.

CO3	Outline the measuring instruments in paper industry		
M3.01	Suggest various process parameters in wood and paper industries	5	Understanding
M3.02	Discuss the properties of paper	5	Understanding
M3.03	Illustrate the working principle for measurement of properties of paper	5	Understanding

Contents:

Definitions and methods of determination of :

Physical properties - Grammage, caliper, bulk, smoothness and porosity;

Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness:

Optical Properties: brightness, opacity, gloss and color:

Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability.

Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper

CO4	Explain the modes of control action and the computer control system		
M4.01	Comprehend the principles of automatic process control.	3	Understanding
M4.02	Comprehend computerized control system.	3	Understanding
M4.03	Summarize final control elements.	4	Understanding
M4.04	Outline instrumented safety system.	4	Understanding
	Series Test - II	1	
Contents: The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks.			

Text / Reference:

T/R	Book Title/Author
T1	Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990.
R1	Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008.
R2	Ghosh A.K., “Introduction to Instrumentation and Control”, 4th Ed., Prentice Hall of India.
R3	Casey J.P., “Pulp and Paper Chemistry and Chemical Technology”, 3rd Ed., Vol 3, John Wiley & Sons.
R4	Robert J’ Measurement and control in paper making’ Mc Grill
R5	Mark R. E., “Handbook of Physical and Mechanical Testing of Paper and Paperboard”, Vol. 1, Marcel Dekker.
R6	Sankaranarayanan, P.E., “Pulp and Paper Industries Technology and Instrumentation”, Kotharis Desk book series, 1995.
R7	G. Stephanopoulos, ‘Chemical Process Control - Theory and Practice’, Prentice Hall of India Ltd.
R8	Lecture notes IPIR, Bangalore

Online Resources:

Sl.No	Website Link
1	https://www.nptel.ac.in/courses/103103037/
2	https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-notes-ics-pdf-notes/